

#1 · Foundations of Adult Cardiac Anesthesiology

Which fetal shunt carries the best-oxygenated blood across the atrial septum, and where does it go?

Answer

The foramen ovale directs inferior vena caval blood into the left atrium, so the brain and coronary arteries receive the most saturated blood.

#2 · Foundations of Adult Cardiac Anesthesiology

What happens at birth to close the foramen ovale and ductus arteriosus?

Answer

Pulmonary resistance falls and left atrial pressure exceeds right, closing the foramen flap. Higher oxygen tension and falling prostaglandin E close the ductus over the first days.

#3 · Foundations of Adult Cardiac Anesthesiology

What keeps a ductal-dependent lesion alive in a newborn?

Answer

Prostaglandin E1 infusion keeps the ductus arteriosus open.

#4 · Foundations of Adult Cardiac Anesthesiology

Which primitive heart segment forms the trabeculated right ventricle and smooth outflow tracts?

Answer

The bulbus cordis.

#5 · Foundations of Adult Cardiac Anesthesiology

From which tissue does the aorticopulmonary septum arise, and what is the failure state?

Answer

Neural crest cells. Failure to septate the truncus gives persistent truncus arteriosus.

#6 · Foundations of Adult Cardiac Anesthesiology

What fraction of people have right coronary dominance, and what defines it?

Answer

About 85 percent. Dominance means the right coronary artery gives the posterior descending artery.

#7 · Foundations of Adult Cardiac Anesthesiology

Why is the posteromedial papillary muscle more prone to infarction and rupture?

Answer

It usually has a single blood supply from the posterior descending artery, while the anterolateral muscle has a dual supply.

#8 · Foundations of Adult Cardiac Anesthesiology

Which artery usually supplies the SA node and AV node?

Answer

The right coronary artery supplies the SA node in about 60 percent and the AV node in about 90 percent.

#9 · Foundations of Adult Cardiac Anesthesiology

Where does most left ventricular venous blood drain, and what is its typical oxygen saturation?

Answer

Into the coronary sinus, about 85 percent of left ventricular flow, with a saturation of about 30 to 40 percent.

#10 · Foundations of Adult Cardiac Anesthesiology

Which nerve hooks under the aortic arch beside the ligamentum arteriosum?

Answer

The left recurrent laryngeal nerve.

#11 · Foundations of Adult Cardiac Anesthesiology

What bounds the triangle of Koch and what lies within it?

Answer

The tendon of Todaro, the tricuspid septal leaflet and the coronary sinus ostium. The AV node lies within it.

#12 · Foundations of Adult Cardiac Anesthesiology

Why is the right main bronchus more prone to receive aspirate or endobronchial tubes?

Answer

It is shorter, wider and more vertical, about 25 degrees from vertical against about 45 degrees on the left.

#13 · Foundations of Adult Cardiac Anesthesiology

What are the five phases of the ventricular myocyte action potential?

Answer

0 sodium influx, 1 transient outward potassium, 2 calcium plateau, 3 potassium repolarization, 4 resting potential of about minus 90 mV set by I_{K1} .

#14 · Foundations of Adult Cardiac Anesthesiology

Which current produces spontaneous phase 4 depolarization in the SA node?

Answer

The funny current (I_f) through HCN channels, aided late by T-type calcium current.

#15 · Foundations of Adult Cardiac Anesthesiology

Why do calcium channel blockers and adenosine slow the AV node but sodium channel blockers do not?

Answer

AV nodal phase 0 depends on L-type calcium current, not fast sodium current.

#16 · Foundations of Adult Cardiac Anesthesiology

How does intravenous calcium help in hyperkalemia with ECG changes?

Answer

It restores the gap between resting and threshold potential and stabilizes the membrane. It does not lower serum potassium.

#17 · Foundations of Adult Cardiac Anesthesiology

On the central venous waveform, what causes a giant cannon a wave?

Answer

Atrial contraction against a closed tricuspid valve, as in junctional rhythm, complete heart block or ventricular pacing without atrial sensing.

#18 · Foundations of Adult Cardiac Anesthesiology

What does a tall v wave suggest on the central venous or wedge tracing?

Answer

Tricuspid regurgitation on the central venous tracing, and mitral regurgitation on the pulmonary capillary wedge tracing.

#19 · Foundations of Adult Cardiac Anesthesiology

On a pressure-volume loop, what does the slope of the end-systolic pressure-volume relationship measure?

Answer

Contractility, as end-systolic elastance, which is relatively load independent.

#20 · Foundations of Adult Cardiac Anesthesiology

How does the pressure-volume loop change with increased afterload?

Answer

It becomes taller and narrower, with a higher end-systolic pressure and a smaller stroke volume.

#21 · Foundations of Adult Cardiac Anesthesiology

State Laplace's law for ventricular wall stress.

Answer

Wall stress equals pressure times radius divided by twice the wall thickness. Dilation raises it and hypertrophy lowers it.

#22 · Foundations of Adult Cardiac Anesthesiology

What are the determinants of myocardial oxygen demand and supply?

Answer

Demand: heart rate, wall tension and contractility.
Supply: coronary perfusion pressure (aortic diastolic minus LVEDP), diastolic time, and arterial oxygen content.

#23 · Foundations of Adult Cardiac Anesthesiology

Why is the right ventricle so sensitive to acute rises in pulmonary vascular resistance?

Answer

It is thin and compliant and cannot acutely generate a mean pulmonary artery pressure much above 40 mmHg, so hypoxia, acidosis or protamine can precipitate right heart failure.

#24 · Foundations of Adult Cardiac Anesthesiology

Calculate PVR when mean PA pressure is 40, wedge is 15 and cardiac output is 5 L/min.

Answer

$$\text{PVR} = 80 \times (40 \text{ minus } 15) \text{ divided by } 5 = 400$$

dyne-sec per cm⁵, or 5 Wood units.

#25 · Foundations of Adult Cardiac Anesthesiology

What is the Bezold-Jarisch reflex and when does it appear?

Answer

Inferoposterior ventricular receptors trigger bradycardia and hypotension, seen with inferior wall reperfusion, hypovolemia and high spinal block.

#26 · Preoperative Evaluation for Cardiac Surgery

Which preoperative ECG finding makes a pulmonary artery catheter risky for complete heart block?

Answer

Preexisting left bundle branch block, because catheter-induced right bundle branch block then produces complete heart block. Have pacing available.

#27 · Preoperative Evaluation for Cardiac Surgery

Why can the ECG not be read for ischemia in a ventricular-paced patient?

Answer

Right ventricular pacing produces a left bundle branch block pattern that masks ST and T changes.

#28 · Preoperative Evaluation for Cardiac Surgery

What RV findings on echo suggest RV dysfunction?

Answer

TAPSE below 17 mm or tricuspid annular S prime below 9.5 cm/s.

#29 · Preoperative Evaluation for Cardiac Surgery

What defines severe aortic stenosis by echo?

Answer

Valve area 1.0 cm² or less, mean gradient 40 mmHg or more, or peak velocity 4 m/s or more.

#30 · Preoperative Evaluation for Cardiac Surgery

What valve area defines severe mitral stenosis in the 2020 ACC/AHA guideline?

Answer

A mitral valve area of 1.5 cm² or less.

#31 · Preoperative Evaluation for Cardiac Surgery

What is the low-flow, low-gradient aortic stenosis pattern?

Answer

A small valve area with a mean gradient under 40 mmHg because stroke volume is low. Stroke volume index is below 35 mL/m² in the paradoxical form.

#32 · Preoperative Evaluation for Cardiac Surgery

What does low-dose dobutamine echo show in true severe aortic stenosis?

Answer

The valve area stays small while flow rises and the gradient climbs. In pseudostenosis the valve area increases.

#33 · Preoperative Evaluation for Cardiac Surgery

What defines contractile reserve on dobutamine echo in low-gradient aortic stenosis?

Answer

A stroke volume increase of 20 percent or more. Its absence predicts higher operative mortality.

#34 · Preoperative Evaluation for Cardiac Surgery

What CT calcium score supports severe aortic stenosis in paradoxical low-flow disease?

Answer

About 2,000 Agatston units or more in men and 1,200 or more in women.

#35 · Preoperative Evaluation for Cardiac Surgery

Which conditions make exercise ECG testing unreliable?

Answer

Left bundle branch block, ventricular pacing, LVH with repolarization changes, digoxin effect, and preexcitation.

#36 · Preoperative Evaluation for Cardiac Surgery

What does a biphasic response on dobutamine stress echo indicate?

Answer

Viable but ischemic myocardium: improvement at low dose and worsening at high dose.

#37 · Preoperative Evaluation for Cardiac Surgery

When are vasodilator stress agents contraindicated?

Answer

Active bronchospasm and high-grade AV block. Caffeine is held 12 to 24 hours because it blocks adenosine receptors.

#38 · Preoperative Evaluation for Cardiac Surgery

Why is PET often better than SPECT?

Answer

Higher spatial resolution and fewer attenuation artifacts, helpful in obese patients and women. FDG PET also identifies hibernating myocardium.

#39 · Preoperative Evaluation for Cardiac Surgery

What does a perfusion-metabolism mismatch on PET predict?

Answer

Hibernating myocardium that is likely to recover function after revascularization.

#40 · Preoperative Evaluation for Cardiac Surgery

What MRI finding predicts that a segment will not recover after revascularization?

Answer

Late gadolinium enhancement through more than 50 percent of wall thickness.

#41 · Preoperative Evaluation for Cardiac Surgery

What angiographic thresholds define significant coronary stenosis?

Answer

50 percent or more in the left main and 70 percent or more elsewhere.

#42 · Preoperative Evaluation for Cardiac Surgery

What FFR and iFR values mark a significant lesion?

Answer

Fractional flow reserve 0.80 or below, instantaneous wave-free ratio 0.89 or below.

#43 · Preoperative Evaluation for Cardiac Surgery

What are normal right-sided pressures?

Answer

RA mean 2 to 6 mmHg, RV 25/0 to 5 mmHg, PA 25/10 with mean 15 mmHg, wedge 6 to 12 mmHg.

#44 · Preoperative Evaluation for Cardiac Surgery

What is the hemodynamic definition of pulmonary hypertension?

Answer

A mean pulmonary artery pressure above 20 mmHg.

#45 · Preoperative Evaluation for Cardiac Surgery

What does a large v wave on the wedge tracing suggest?

Answer

Mitral regurgitation, though it also occurs with ventricular septal defect or a noncompliant left atrium.

#46 · Preoperative Evaluation for Cardiac Surgery

Why does wedge pressure overestimate LV end-diastolic pressure in mitral stenosis?

Answer

Wedge reflects left atrial pressure and a diastolic gradient exists across the stenotic valve.

#47 · Preoperative Evaluation for Cardiac Surgery

What causes cannon a waves?

Answer

Atrial contraction against a closed AV valve, as in complete heart block or junctional rhythm.

#48 · Preoperative Evaluation for Cardiac Surgery

What is the pressure pattern of constrictive pericarditis?

Answer

Prominent y descent, dip and plateau (square root sign), diastolic equalization, and ventricular interdependence.

#49 · Preoperative Evaluation for Cardiac Surgery

**What is the
Brockenbrough-Braunwald-Morrow sign?**

Answer

In hypertrophic obstructive cardiomyopathy, aortic pulse pressure falls after a premature beat because the obstruction deepens.

#50 · Preoperative Evaluation for Cardiac Surgery

**How is pulmonary vascular resistance
calculated?**

Answer

(Mean PA pressure minus wedge pressure) divided by cardiac output. Example: $(35 \text{ minus } 15) / 5 = 4$ Wood units, or 320 dyn.s/cm⁵.

#51 · Preoperative Evaluation for Cardiac Surgery

How is Qp/Qs calculated from saturations?

Answer

(Systemic arterial minus mixed venous saturation) divided by (pulmonary venous minus pulmonary arterial saturation).

#52 · Preoperative Evaluation for Cardiac Surgery

What Qp/Qs generally supports shunt closure?

Answer

1.5 or more, together with other criteria. Reversal with a ratio below 1.0 marks Eisenmenger physiology.

#53 · Preoperative Evaluation for Cardiac Surgery

What saturation step-up localizes a shunt?

Answer

A rise of more than 7 percent between chambers, for example at the atrial level in an atrial septal defect.

#54 · Preoperative Evaluation for Cardiac Surgery

How is cardiac output derived by the Fick principle?

Answer

Oxygen consumption divided by the arteriovenous oxygen content difference, with content $1.34 \times \text{Hb} \times \text{saturation}$. VO_2 is often assumed at 125 mL/min/m².

#55 · Preoperative Evaluation for Cardiac Surgery

What is the Hakki estimate of valve area?

Answer

Cardiac output in L/min divided by the square root of the gradient in mmHg.

#56 · Preoperative Evaluation for Cardiac Surgery

How should chronic beta-blockers be handled before cardiac surgery?

Answer

Continue through the morning of surgery. Do not start one on the day of surgery in a naive patient, per POISE.

#57 · Preoperative Evaluation for Cardiac Surgery

What is the ARNI washout rule?

Answer

Sacubitril-valsartan needs a 36-hour gap from any ACE inhibitor because of angioedema risk.

#58 · Preoperative Evaluation for Cardiac Surgery

Why are SGLT2 inhibitors held before surgery, and for how long?

Answer

They cause euglycemic ketoacidosis. Hold 3 days, or 4 days for ertugliflozin.

#59 · Preoperative Evaluation for Cardiac Surgery

How long are P2Y12 inhibitors held before elective cardiac surgery?

Answer

Clopidogrel at least 5 days, ticagrelor at least 3 days, prasugrel at least 7 days. Aspirin is continued.

#60 · Preoperative Evaluation for Cardiac Surgery

How are DOACs and warfarin handled before surgery?

Answer

Warfarin stopped about 5 days to INR near 1.5 or lower. Apixaban, rivaroxaban and edoxaban held at least 48 hours. DOACs are not used with mechanical valves.

#61 · Intraoperative Monitoring

Which single ECG lead is most sensitive for detecting anterior and lateral ischemia?

Answer

Lead V5 detects roughly 75 percent of ischemic events. Leads II plus V5 together detect roughly 80 percent.

#62 · Intraoperative Monitoring

Which ECG filter mode should be used for ST segment analysis?

Answer

The diagnostic filter mode. The monitor mode filter distorts the ST segment.

#63 · Intraoperative Monitoring

Name four conditions that make ECG ST analysis unreliable.

Answer

Left bundle branch block, ventricular pacing, electrocautery noise, and digoxin effect (scooped ST depression).

#64 · Intraoperative Monitoring

Why trust the arterial waveform over the monitor heart rate with a pacemaker?

Answer

The ECG monitor can count pacer spikes or T waves, giving a rate far above the true pulse rate. The pulse waveform shows actual perfusion rate.

#65 · Intraoperative Monitoring

What does an underdamped arterial line do to the pressure readings?

Answer

It overestimates systolic and underestimates diastolic pressure. The mean pressure stays roughly correct.

#66 · Intraoperative Monitoring

What does an overdamped arterial line do to the trace?

Answer

It flattens the waveform and underestimates systolic pressure, with a roughly correct mean. Causes are air bubbles, kinks, or clot.

#67 · Intraoperative Monitoring

How does the fast flush test show optimal damping?

Answer

One or two oscillations after the flush before the trace settles. Many oscillations means underdamped; slow return without overshoot means overdamped.

#68 · Intraoperative Monitoring

Why can radial pressure mislead after separation from bypass?

Answer

Radial systolic pressure can run 10 to 30 mmHg below central aortic pressure. Check a femoral or aortic pressure before starting a vasopressor.

#69 · Intraoperative Monitoring

Why is the right radial line misleading with right axillary cannulation?

Answer

The right radial reads high from direct flow in the cannulated branch, so a left radial or femoral line is added. Systemic pressure is not reflected.

#70 · Intraoperative Monitoring

What does pulse pressure variation measure?

Answer

The maximum minus minimum pulse pressure over a respiratory cycle divided by their mean. Above 12 to 13 percent predicts fluid responsiveness.

#71 · Intraoperative Monitoring

List the conditions required for pulse pressure variation to be valid.

Answer

Controlled ventilation, tidal volume of at least 8 mL/kg predicted body weight, regular sinus rhythm, closed chest, normal lung compliance, no right ventricular failure.

#72 · Intraoperative Monitoring

Which first-choice vein is used for central venous access in cardiac surgery?

Answer

The right internal jugular vein, placed with real-time ultrasound and the Seldinger technique.

#73 · Intraoperative Monitoring

Why is the subclavian vein avoided in coagulopathy?

Answer

It cannot be compressed if the artery is punctured, and it carries a higher pneumothorax risk, though its infection rate is lower.

#74 · Intraoperative Monitoring

Where should a central venous catheter tip rest?

Answer

At the cavoatrial junction, above the pericardial reflection.

#75 · Intraoperative Monitoring

What produces the a wave of the CVP trace?

Answer

Atrial contraction, following the P wave. It is absent in atrial fibrillation.

#76 · Intraoperative Monitoring

What causes cannon a waves on the CVP trace?

Answer

Atrial contraction against a closed tricuspid valve, as in junctional rhythm, ventricular pacing, or complete heart block.

#77 · Intraoperative Monitoring

What is the CVP waveform finding in tricuspid regurgitation?

Answer

A large systolic v wave, with the c wave and x descent merged.

#78 · Intraoperative Monitoring

Which CVP descent is lost in cardiac tamponade?

Answer

The y descent, because the ventricle cannot fill in early diastole.

#79 · Intraoperative Monitoring

What are normal PA catheter pressures?

Answer

Right ventricle 25/0 to 8, pulmonary artery 25/8 with mean about 15, wedge 6 to 12 mmHg, right atrium mean 2 to 8 mmHg.

#80 · Intraoperative Monitoring

How much air does the pulmonary artery catheter balloon hold, and when is it inflated?

Answer

1.5 mL. Inflate only in the right atrium on the way in, and deflate when withdrawing.

#81 · Intraoperative Monitoring

Why can the wedge pressure overestimate left ventricular filling?

Answer

Mitral stenosis, high airway pressure or PEEP, and pulmonary venous disease raise it. A stiff left ventricle makes it underestimate end-diastolic pressure.

#82 · Intraoperative Monitoring

What does a large v wave on the wedge tracing indicate?

Answer

Mitral regurgitation, or any lesion raising left atrial filling in systole such as a ventricular septal defect.

#83 · Intraoperative Monitoring

In which West lung zone must the catheter tip lie for a valid wedge?

Answer

Zone 3, where pulmonary venous pressure exceeds alveolar pressure.

#84 · Intraoperative Monitoring

What are the risk factors for pulmonary artery rupture from a PA catheter?

Answer

Age over 60, pulmonary hypertension, hypothermia and rewarming on bypass, anticoagulation, distal migration, and balloon overinflation.

#85 · Intraoperative Monitoring

What should be done with the PA catheter before cooling on bypass?

Answer

Withdraw it 2 to 3 cm, because the tip migrates distally during cooling.

#86 · Intraoperative Monitoring

How does thermodilution cardiac output relate to the temperature curve?

Answer

Output is inversely related to the area under the curve. Low output gives a large slow curve.

#87 · Intraoperative Monitoring

Which conditions make thermodilution cardiac output unreliable?

Answer

Severe tricuspid regurgitation and intracardiac shunts.

#88 · Intraoperative Monitoring

What is a normal mixed venous oxygen saturation, and what lowers it?

Answer

65 to 75 percent. Low cardiac output, anemia, hypoxemia, and high oxygen consumption lower it.

#89 · Intraoperative Monitoring

What does esophageal Doppler measure, and what does it assume?

Answer

Descending thoracic aortic blood velocity. It assumes about 70 percent of cardiac output flows through the descending aorta.

#90 · Intraoperative Monitoring

Which transgastric view monitors volume status and wall motion in all coronary territories?

Answer

The transgastric midpapillary short-axis view.

#91 · Intraoperative Monitoring

How is stroke volume calculated by echo, and what is the main source of error?

Answer

LVOT area times LVOT velocity time integral. The outflow tract diameter is squared, so its error dominates.

#92 · Intraoperative Monitoring

What does global longitudinal strain show, and what is its normal value?

Answer

Percent myocardial shortening by speckle tracking. Normal is about negative 20 percent; a less negative value means worse function.

#93 · Intraoperative Monitoring

What NIRS change triggers action during cardiac surgery?

Answer

A fall of more than 20 percent from baseline, or an absolute rSO₂ below 50 percent.

#94 · Intraoperative Monitoring

Which agents suppress motor evoked potentials?

Answer

Neuromuscular blockers and volatile anesthetics above 0.5 MAC, so a total intravenous technique is usually used.

#95 · Intraoperative Monitoring

What cerebrospinal fluid pressure is the usual drain target in thoracoabdominal aortic repair?

Answer

10 to 12 mmHg.

#96 · Intraoperative Monitoring

What gradient limits are used when rewarming on bypass?

Answer

Arterial inflow no higher than 37 degrees Celsius, and an inflow to venous return gradient of no more than about 10 degrees Celsius.

#97 · Intraoperative Monitoring

What is the normal ACT, and the usual target for full bypass?

Answer

Normal is about 80 to 120 seconds. Bypass target is above 480 seconds, with 400 to 480 accepted in some centers.

#98 · Intraoperative Monitoring

What is the initial heparin dose before cannulation?

Answer

300 to 400 units/kg, with an ACT check 3 to 5 minutes later.

#99 · Intraoperative Monitoring

What is the commonest cause of heparin resistance and its treatment?

Answer

Acquired antithrombin deficiency from preoperative heparin infusion. Treat with antithrombin concentrate or fresh frozen plasma.

#100 · Intraoperative Monitoring

What does a long TEG R time that corrects with heparinase indicate?

Answer

Residual heparin effect, which calls for more protamine.

#101 · Management of Cardiopulmonary Bypass

Usual heparin dose for CPB?

Answer

300 to 400 units/kg IV before cannulation.

#102 · Management of Cardiopulmonary Bypass

Target ACT for full CPB?

Answer

Above 480 seconds, with many centers accepting 400 to 480 seconds.

#103 · Management of Cardiopulmonary Bypass

Normal baseline ACT?

Answer

About 80 to 120 seconds.

#104 · Management of Cardiopulmonary Bypass

How often is the ACT checked on bypass?

Answer

About every 30 minutes, with supplemental heparin when below target.

#105 · Management of Cardiopulmonary Bypass

What prolongs the ACT without a heparin effect?

Answer

Hypothermia, hemodilution, thrombocytopenia and platelet inhibitors.

#106 · Management of Cardiopulmonary Bypass

How does aprotinin affect ACT assays?

Answer

It prolongs the celite ACT much more than the kaolin ACT, so a celite target near 750 seconds or a kaolin test is used.

#107 · Management of Cardiopulmonary Bypass

Mechanism of heparin?

Answer

It binds antithrombin and speeds its inactivation of thrombin and factor Xa about 1,000 fold.

#108 · Management of Cardiopulmonary Bypass

Half-life of heparin?

Answer

About 60 to 90 minutes, dose dependent.

#109 · Management of Cardiopulmonary Bypass

Usual protamine dose to reverse heparin?

Answer

1 to 1.3 mg per 100 units of total heparin, or a dose from a measured heparin concentration.

#110 · Management of Cardiopulmonary Bypass

How should protamine be given?

Answer

Slowly over 10 minutes or more, off bypass, to avoid hypotension.

#111 · Management of Cardiopulmonary Bypass

What is a type I protamine reaction?

Answer

Hypotension from rapid injection, caused by histamine-like vasodilation.

#112 · Management of Cardiopulmonary Bypass

What is a type III protamine reaction?

Answer

Catastrophic pulmonary vasoconstriction with a high PA pressure, acute right heart failure and systemic hypotension.

#113 · Management of Cardiopulmonary Bypass

Risk factors for serious protamine reactions?

Answer

Prior protamine exposure, NPH insulin, fish allergy and prior vasectomy.

#114 · Management of Cardiopulmonary Bypass

What is heparin rebound?

Answer

Return of anticoagulation hours after reversal as heparin leaves tissue stores; treat with a small protamine dose guided by the ACT.

#115 · Management of Cardiopulmonary Bypass

Define heparin resistance.

Answer

Failure to reach an ACT of 480 seconds after 400 to 600 units/kg of heparin.

#116 · Management of Cardiopulmonary Bypass

Most common cause of heparin resistance?

Answer

Acquired antithrombin deficiency after several days of continuous heparin infusion.

#117 · Management of Cardiopulmonary Bypass

Treatment of heparin resistance after extra heparin fails?

Answer

Antithrombin concentrate, commonly 500 to 1,000 units, or fresh frozen plasma if concentrate is unavailable.

#118 · Management of Cardiopulmonary Bypass

What is the mechanism of type II HIT?

Answer

IgG antibodies to platelet factor 4 bound to heparin activate platelets through the Fc receptor.

#119 · Management of Cardiopulmonary Bypass

Typical timing of HIT?

Answer

Platelet fall of more than 50 percent 5 to 10 days after starting heparin, or within hours after recent exposure in the last 30 days.

#120 · Management of Cardiopulmonary Bypass

What is the 4Ts score?

Answer

A clinical probability tool: thrombocytopenia, timing, thrombosis and other causes of thrombocytopenia.

#121 · Management of Cardiopulmonary Bypass

Which tests diagnose HIT?

Answer

Anti-PF4 ELISA to screen and the serotonin release assay to confirm.

#122 · Management of Cardiopulmonary Bypass

Preferred heparin alternative for CPB in acute HIT?

Answer

Bivalirudin, a direct thrombin inhibitor.

#123 · Management of Cardiopulmonary Bypass

Bivalirudin dosing for CPB?

Answer

1 mg/kg bolus, 50 mg in the pump prime, then 2.5 mg/kg/h infusion.

#124 · Management of Cardiopulmonary Bypass

ACT target on bivalirudin?

Answer

Above 2.5 times baseline, commonly over 400 to 500 seconds.

#125 · Management of Cardiopulmonary Bypass

How is bivalirudin cleared?

Answer

Plasma enzymatic proteolysis plus about 20 percent renal clearance; half-life near 25 minutes.

#126 · Management of Cardiopulmonary Bypass

Main hazard of bivalirudin on CPB?

Answer

Clot formation in areas of stasis because thrombin stays active; keep blood circulating and avoid stagnant reservoirs.

#127 · Management of Cardiopulmonary Bypass

Is there a reversal agent for bivalirudin?

Answer

No.

#128 · Management of Cardiopulmonary Bypass

Why is argatroban a poor choice for CPB?

Answer

Hepatic clearance and a half-life near 45 minutes make it hard to titrate and wear off.

#129 · Management of Cardiopulmonary Bypass

How is remote HIT managed for CPB?

Answer

With negative antibodies and an event over 100 days ago, a single brief heparin exposure during CPB is accepted, with none before or after.

#130 · Management of Cardiopulmonary Bypass

What do you do with elective surgery in active HIT?

Answer

Delay until antibodies are gone and avoid all heparin including flushes and coated catheters.

#131 · Management of Cardiopulmonary Bypass

Mechanism of tranexamic acid?

Answer

A lysine analog that blocks plasminogen lysine-binding sites so it cannot attach to fibrin.

#132 · Management of Cardiopulmonary Bypass

Major adverse effect of high-dose TXA?

Answer

Postoperative convulsive seizures through glycine and GABA-A receptor inhibition.

#133 · Management of Cardiopulmonary Bypass

Who has higher seizure risk with TXA?

Answer

Older patients, renal impairment and open-chamber surgery.

#134 · Management of Cardiopulmonary Bypass

Why was aprotinin withdrawn in 2008?

Answer

The BART trial showed higher mortality than lysine analogs, with renal injury risk.

#135 · Management of Cardiopulmonary Bypass

Basic components of the CPB circuit?

Answer

Venous reservoir, pump, oxygenator with heat exchanger, arterial line filter and cannulae.

#136 · Management of Cardiopulmonary Bypass

Key behavior of a centrifugal pump versus a roller pump?

Answer

Centrifugal flow falls as afterload rises and it cannot pump large air volumes; a roller pump is occlusive and can generate dangerous pressure.

#137 · Management of Cardiopulmonary Bypass

Calculate hematocrit on bypass: blood volume 5 L, Hct 40 percent, prime 1.5 L.

Answer

Red cell volume 2 L divided by 6.5 L total gives about 31 percent.

#138 · Management of Cardiopulmonary Bypass

Lowest acceptable hematocrit on bypass?

Answer

Usually 21 to 24 percent; lower values are linked to acute kidney injury.

#139 · Management of Cardiopulmonary Bypass

Standard central cannulation sites?

Answer

Distal ascending aorta for arterial and right atrial appendage two-stage cannula for venous drainage.

#140 · Management of Cardiopulmonary Bypass

When is bicaval cannulation used?

Answer

When the right heart will be opened, with caval snares.

#141 · Management of Cardiopulmonary Bypass

Signs of aortic dissection at CPB onset?

Answer

Sudden rise in line pressure, loss of pulsation, blue aortic root, falling venous return and unequal arterial pressures.

#142 · Management of Cardiopulmonary Bypass

Response to suspected dissection on cannulation?

Answer

Reduce or stop flow, move to another cannulation site such as femoral or axillary, and repair urgently.

#143 · Management of Cardiopulmonary Bypass

Effect of arterial cannula tip in the innominate artery?

Answer

High right radial pressure, low left radial pressure and cerebral hyperperfusion.

#144 · Management of Cardiopulmonary Bypass

Best tool to detect aortic atheroma before cannulation?

Answer

Epiaortic ultrasound, more sensitive than palpation.

#145 · Management of Cardiopulmonary Bypass

Preferred arterial site for dissection or arch surgery?

Answer

Right axillary artery through an 8 mm graft, allowing antegrade flow and selective cerebral perfusion.

#146 · Management of Cardiopulmonary Bypass

Risk of femoral arterial cannulation?

Answer

Retrograde embolization of aortic atheroma and malperfusion of the true lumen in dissection.

#147 · Management of Cardiopulmonary Bypass

Why monitor a second arterial line with axillary cannulation?

Answer

The right radial line reads a cannula-affected pressure, so a left radial or femoral line shows true systemic pressure.

#148 · Management of Cardiopulmonary Bypass

Causes of poor venous drainage on CPB?

Answer

Hypovolemia, kinked or malpositioned cannula, venous air lock or a lifted heart.

#149 · Management of Cardiopulmonary Bypass

Main mechanism of hyperkalemic cardioplegia?

Answer

Potassium of 16 to 30 mEq/L depolarizes the membrane, inactivates sodium channels and arrests the heart in diastole.

#150 · Management of Cardiopulmonary Bypass

Usual blood to crystalloid ratio in standard blood cardioplegia?

Answer

4 parts blood to 1 part crystalloid.

#151 · Management of Cardiopulmonary Bypass

What is del Nido cardioplegia?

Answer

1 part blood to 4 parts crystalloid with lidocaine and magnesium, one 20 mL/kg dose lasting 60 to 90 minutes.

#152 · Management of Cardiopulmonary Bypass

Target temperature of cold cardioplegia?

Answer

4 to 10 degrees C.

#153 · Management of Cardiopulmonary Bypass

Typical antegrade root pressure for cardioplegia?

Answer

70 to 100 mmHg.

#154 · Management of Cardiopulmonary Bypass

Retrograde cardioplegia pressure target?

Answer

30 to 40 mmHg in the coronary sinus to avoid sinus rupture.

#155 · Management of Cardiopulmonary Bypass

Why does antegrade cardioplegia fail in significant aortic regurgitation?

Answer

Solution leaks into the left ventricle, causing distension and poor myocardial delivery; use ostial or retrograde delivery.

#156 · Management of Cardiopulmonary Bypass

Weakness of retrograde cardioplegia?

Answer

Poor protection of the right ventricle and posterior septum.

#157 · Management of Cardiopulmonary Bypass

What does return of electrical activity on the cross-clamp mean?

Answer

Cardioplegia is washing out or the heart is rewarming, so redose.

#158 · Management of Cardiopulmonary Bypass

Typical normothermic CPB flow?

Answer

2.2 to 2.4 L/min/m².

#159 · Management of Cardiopulmonary Bypass

Common MAP target on CPB?

Answer

50 to 80 mmHg, higher in cerebrovascular disease, chronic hypertension or carotid stenosis.

#160 · Management of Cardiopulmonary Bypass

Why does MAP fall at CPB onset?

Answer

Hemodilution lowers viscosity, pulsatile flow is lost and circulating catecholamines are diluted.

#161 · Management of Cardiopulmonary Bypass

DO₂i goal to reduce kidney injury on CPB?

Answer

Above 272 mL/min/m².

#162 · Management of Cardiopulmonary Bypass

Target mixed venous saturation on CPB?

Answer

Above 65 to 70 percent.

#163 · Management of Cardiopulmonary Bypass

Treatment of vasoplegia after CPB?

Answer

Vasopressin, then methylene blue 1.5 to 2 mg/kg, which inhibits guanylate cyclase and nitric oxide signaling.

#164 · Management of Cardiopulmonary Bypass

Risk factors for post-CPB vasoplegia?

Answer

ACE inhibitors, ARBs, amiodarone, low ejection fraction and long bypass time.

#165 · Management of Cardiopulmonary Bypass

Effect of hypothermia on cerebral metabolic rate?

Answer

Falls about 6 to 7 percent per degree C.

#166 · Management of Cardiopulmonary Bypass

Maximum arterial inflow temperature during rewarming?

Answer

37 degrees C, to avoid cerebral hyperthermia.

#167 · Management of Cardiopulmonary Bypass

Rewarming rate and gradient limits?

Answer

About 0.5 degrees C per minute or slower, with water to blood gradient under 10 degrees C.

#168 · Management of Cardiopulmonary Bypass

Why do cold agglutinins matter for CPB?

Answer

IgM antibodies agglutinate red cells below 37 degrees C, causing hemolysis and poor myocardial protection.

#169 · Management of Cardiopulmonary Bypass

Strategy for cold agglutinin patients?

Answer

Warm blood cardioplegia, avoid systemic hypothermia, warm the room and fluids, and consider preoperative plasmapheresis if symptomatic with a high titer.

#170 · Management of Cardiopulmonary Bypass

Define alpha-stat management.

Answer

Uncorrected blood gas values at 37 degrees C, targeting pH 7.40 and PaCO₂ 40 mmHg, preserving cerebral autoregulation.

#171 · Management of Cardiopulmonary Bypass

Define pH-stat management.

Answer

Gas values corrected to the patient's temperature, adding CO₂ to hold pH 7.40, which raises cerebral blood flow and abolishes autoregulation.

#172 · Management of Cardiopulmonary Bypass

Which acid-base strategy is usual in adults?

Answer

Alpha-stat.

#173 · Management of Cardiopulmonary Bypass

Why is awareness a risk during rewarming?

Answer

Vasodilation and falling drug concentrations lighten the anesthetic, so add volatile agent or a hypnotic and monitor processed EEG.

#174 · Management of Cardiopulmonary Bypass

Systemic inflammatory mediators activated by CPB?

Answer

Complement C3a and C5a, neutrophils and cytokines such as IL-6 and TNF-alpha.

#175 · Management of Cardiopulmonary Bypass

Glucose threshold for insulin during CPB?

Answer

Treat glucose above 180 mg/dL, partly to reduce neurologic injury.

#176 · Mechanical Circulatory Support

Where should the IABP tip sit?

Answer

1 to 2 cm distal to the origin of the left subclavian artery, near the carina on chest radiograph, above the renal arteries.

#177 · Mechanical Circulatory Support

When does the IABP inflate?

Answer

At the dicrotic notch, which marks aortic valve closure, in diastole.

#178 · Mechanical Circulatory Support

When does the IABP deflate?

Answer

Just before systole, which lowers aortic volume and afterload before ejection.

#179 · Mechanical Circulatory Support

What are the two hemodynamic effects of the IABP?

Answer

Diastolic augmentation that improves coronary perfusion pressure, and systolic unloading that lowers afterload and myocardial oxygen demand.

#180 · Mechanical Circulatory Support

How much does the IABP raise cardiac output?

Answer

Only about 10 to 20 percent, and it needs an ejecting ventricle and a regular rhythm.

#181 · Mechanical Circulatory Support

What is the effect of early IABP inflation?

Answer

Inflation before aortic valve closure causes premature valve closure, lower stroke volume, higher LV end-diastolic pressure and possibly worse aortic regurgitation.

#182 · Mechanical Circulatory Support

What is the effect of late IABP deflation?

Answer

The balloon is still inflated as ejection starts, raising afterload and oxygen demand and reducing output. It is the most harmful timing error.

#183 · Mechanical Circulatory Support

Name the absolute contraindications to the IABP.

Answer

Significant aortic regurgitation and aortic dissection. Uncontrolled sepsis and bleeding are other contraindications.

#184 · Mechanical Circulatory Support

What did IABP-SHOCK II show?

Answer

No 30-day mortality benefit from routine IABP use in infarct-related cardiogenic shock. The IABP still has a role for mechanical complications of infarction.

#185 · Mechanical Circulatory Support

What do blood or rust-brown flakes in the IABP gas line mean?

Answer

Balloon rupture. Stop pumping and remove or exchange the balloon promptly.

#186 · Mechanical Circulatory Support

Where does an Impella sit and what does it do?

Answer

Across the aortic valve, drawing blood from the LV and ejecting it into the ascending aorta, directly unloading the LV.

#187 · Mechanical Circulatory Support

Approximate maximal flows of Impella 2.5, CP and 5.5?

Answer

About 2.5 L/min, 3.5 to 4 L/min, and up to about 5.5 L/min respectively.

#188 · Mechanical Circulatory Support

Which Impella models need a surgical cutdown?

Answer

The 5.0 and 5.5, usually through the axillary artery. The 2.5 and CP are placed percutaneously through the femoral artery.

#189 · Mechanical Circulatory Support

Where should the Impella CP inlet sit?

Answer

About 3.5 cm below the aortic annulus, in the LV cavity clear of the mitral apparatus and septum.

#190 · Mechanical Circulatory Support

Name the contraindications to an Impella.

Answer

Mechanical aortic valve, LV thrombus, severe aortic stenosis, more than mild aortic regurgitation, and a VSD.

#191 · Mechanical Circulatory Support

What causes Impella suction events and how are they treated?

Answer

Low LV volume or a malpositioned inlet. Give volume, reduce speed, reposition under TEE and treat RV failure.

#192 · Mechanical Circulatory Support

What did DanGer Shock report?

Answer

Lower 180-day mortality with Impella CP in selected patients with STEMI and cardiogenic shock, with more vascular and bleeding complications.

#193 · Mechanical Circulatory Support

Describe TandemHeart cannulation.

Answer

About 21 Fr inflow cannula from the femoral vein across the septum into the left atrium, with a centrifugal pump returning to a femoral artery.

#194 · Mechanical Circulatory Support

Why can a TandemHeart be used with a mechanical aortic valve?

Answer

It does not cross the aortic valve, so the valve and LV thrombus are not a barrier.

#195 · Mechanical Circulatory Support

What is the main hemodynamic downside of the TandemHeart?

Answer

It adds LV afterload because blood returns to the aorta against the ventricle. It does lower left atrial and wedge pressure.

#196 · Mechanical Circulatory Support

What device supports the right ventricle with a dual-lumen right atrium to pulmonary artery cannula?

Answer

The ProtekDuo. The Impella RP also supports the RV, with inflow in the inferior vena cava and outflow in the pulmonary artery.

#197 · Mechanical Circulatory Support

What are the four LVAD strategies?

Answer

Bridge to transplant, destination therapy, bridge to candidacy and bridge to recovery.

#198 · Mechanical Circulatory Support

Which durable LVAD is current, and what is distinctive?

Answer

The HeartMate 3, a centrifugal pump with a fully magnetically levitated rotor and a built-in artificial pulse every two seconds. The HVAD was withdrawn in 2021.

#199 · Mechanical Circulatory Support

How common is RV failure after LVAD and what predicts it?

Answer

About 10 to 40 percent. Low RV stroke work index, CVP to wedge ratio above about 0.63, severe RV dysfunction and high PVR predict it.

#200 · Mechanical Circulatory Support

Why must a PFO be closed before LVAD start?

Answer

LVAD unloading drops left atrial pressure below right atrial pressure, causing right-to-left shunting, hypoxemia and paradoxical emboli.

#201 · Mechanical Circulatory Support

How much aortic regurgitation is acceptable with an LVAD?

Answer

No more than mild. Greater AR forms a closed loop of regurgitant flow and low effective systemic output, so it must be repaired.

#202 · Mechanical Circulatory Support

Why does air after LVAD start cause RV failure?

Answer

Air collects in the anterior right coronary artery and causes RV ischemia. Careful de-airing under TEE prevents it.

#203 · Mechanical Circulatory Support

Which way should the LVAD inflow cannula point?

Answer

Toward the mitral valve, away from the septum and free wall.

#204 · Mechanical Circulatory Support

What TEE findings suggest LVAD speed is too high?

Answer

A small slit-like LV, septum shifted left, suction events and ventricular arrhythmias.

#205 · Mechanical Circulatory Support

What are the four LVAD controller parameters?

Answer

Speed in rpm, estimated flow in L/min, power in watts, and pulsatility index.

#206 · Mechanical Circulatory Support

What do low flow, low pulsatility index and suction alarms suggest on an LVAD?

Answer

Low preload from hypovolemia, tamponade or RV failure. Check volume and RV function before raising speed.

#207 · Mechanical Circulatory Support

What pattern suggests LVAD pump thrombosis?

Answer

A sustained rise in power with hemolysis, dark urine and high LDH.

#208 · Mechanical Circulatory Support

What mean arterial pressure target applies to an LVAD patient?

Answer

About 70 to 80 mmHg and generally no higher than 90, because the pump is sensitive to afterload.

#209 · Mechanical Circulatory Support

What is differential hypoxemia on VA ECMO?

Answer

In peripheral VA ECMO with a recovered LV and poor lungs, the heart delivers poorly oxygenated blood to the coronaries, brain and right arm. Monitor the right radial gas.

#210 · Mechanical Circulatory Support

What happens to the LV on peripheral VA ECMO and how is it vented?

Answer

Afterload rises and a poorly ejecting LV can distend. Use an inotrope, IABP, Impella, surgical vent or atrial septostomy.

#211 · Mechanical Circulatory Support

Name criteria for starting VV ECMO.

Answer

PaO₂/FiO₂ below 80 mmHg for over 6 hours or below 50 mmHg for over 3 hours, or pH below 7.25 with PaCO₂ 60 mmHg or more for over 6 hours.

#212 · Mechanical Circulatory Support

What is recirculation on VV ECMO?

Answer

Oxygenated return blood is drawn back into the drainage port, shown by a high circuit venous saturation with persistent patient hypoxemia.

#213 · Mechanical Circulatory Support

What controls oxygen delivery versus CO₂ removal on ECMO?

Answer

Blood flow and hemoglobin set oxygen delivery, while sweep gas flow mainly sets CO₂ removal.

#214 · Mechanical Circulatory Support

What anticoagulation targets and alternative apply on ECMO?

Answer

Heparin with ACT about 180 to 220 seconds or anti-Xa 0.3 to 0.7 units/mL. Use bivalirudin for HIT.

#215 · Mechanical Circulatory Support

What predicts successful VA ECMO weaning?

Answer

At flow 1 to 1.5 L/min: LVEF above about 20 to 25 percent, aortic VTI above about 10 cm and lateral mitral annular velocity above about 6 cm/s.

#216 · Cardiac Diseases

What is the typical ejection fraction in dilated cardiomyopathy?

Answer

Below 40 percent, with a dilated left ventricle on the flat part of the Starling curve.

#217 · Cardiac Diseases

Which gene is the most common known cause of familial dilated cardiomyopathy?

Answer

Titin truncating variants.

#218 · Cardiac Diseases

How does late gadolinium enhancement pattern separate ischemic from non-ischemic cardiomyopathy?

Answer

Ischemic disease enhances subendocardially in a coronary territory; non-ischemic disease is typically mid-wall.

#219 · Cardiac Diseases

Define peripartum cardiomyopathy.

Answer

Heart failure with ejection fraction below 45 percent in the last month of pregnancy or within 5 months of delivery.

#220 · Cardiac Diseases

Anesthetic goals in dilated cardiomyopathy?

Answer

Preserve contractility, modest preload, afterload reduction, avoid tachycardia and depressants, and have inotropes ready.

#221 · Cardiac Diseases

What inheritance and gene types cause hypertrophic cardiomyopathy?

Answer

Autosomal dominant sarcomere gene variants, mostly beta-myosin heavy chain and myosin-binding protein C.

#222 · Cardiac Diseases

What causes dynamic LVOT obstruction in HCM?

Answer

Systolic anterior motion of the mitral leaflet, pulled into the outflow tract by Venturi and drag forces.

#223 · Cardiac Diseases

Which Doppler signal suggests dynamic LVOT obstruction?

Answer

A late-peaking, dagger-shaped continuous-wave profile.

#224 · Cardiac Diseases

What three changes worsen HCM obstruction?

Answer

Decreased preload, decreased afterload, and increased contractility or heart rate.

#225 · Cardiac Diseases

Preferred vasopressor for hypotension in obstructive HCM?

Answer

Phenylephrine or vasopressin, with volume; avoid epinephrine, dopamine and other inotropes.

#226 · Cardiac Diseases

Resting or provoked LVOT gradient that supports septal reduction therapy?

Answer

50 mmHg or more with persistent symptoms despite maximally tolerated medical therapy.

#227 · Cardiac Diseases

Main complication of alcohol septal ablation?

Answer

Conduction block, including complete heart block needing a pacemaker, from septal infarction via the first septal perforator.

#228 · Cardiac Diseases

How do mavacamten and aficamten work?

Answer

They are cardiac myosin inhibitors that reduce actin-myosin cross bridges and lower the LVOT gradient, and may lower ejection fraction.

#229 · Cardiac Diseases

Hemodynamic hallmark of restrictive cardiomyopathy?

Answer

A stiff ventricle with high filling pressures, small cavity, biatrial enlargement and fixed stroke volume.

#230 · Cardiac Diseases

Why is bradycardia poorly tolerated in restrictive cardiomyopathy?

Answer

Stroke volume is fixed, so cardiac output depends on heart rate.

#231 · Cardiac Diseases

Echo pattern of Takotsubo cardiomyopathy?

Answer

Apical ballooning with basal hyperkinesis after stress, which may cause LVOT obstruction.

#232 · Cardiac Diseases

How is heart failure classified by ejection fraction?

Answer

HFrEF is 40 percent or less, HFmrEF 41 to 49 percent, HFpEF 50 percent or more with elevated filling pressure.

#233 · Cardiac Diseases

Most common etiology of HFrEF?

Answer

Ischemic heart disease.

#234 · Cardiac Diseases

BNP and NT-proBNP cutoffs for acute heart failure?

Answer

BNP above 100 pg/mL or NT-proBNP above 300 pg/mL, with higher age-adjusted cutoffs for older adults.

#235 · Cardiac Diseases

Why use NT-proBNP rather than BNP on sacubitril-valsartan?

Answer

Neprilysin inhibition raises BNP but not NT-proBNP.

#236 · Cardiac Diseases

Common causes of HFpEF?

Answer

Hypertension, obesity, diabetes, atrial fibrillation, aging and infiltrative disease.

#237 · Cardiac Diseases

Echo criteria supporting diastolic dysfunction?

Answer

Septal e prime below 7 cm/s, average E/e prime above 14, left atrial volume index above 34 mL/m², TR velocity above 2.8 m/s.

#238 · Cardiac Diseases

Mitral inflow pattern in grade 1 diastolic dysfunction?

Answer

Impaired relaxation with E/A ratio below 0.8.

#239 · Cardiac Diseases

Four pillars of guideline-directed therapy for HFrEF?

Answer

ARNI (or ACE inhibitor or ARB), evidence-based beta blocker, mineralocorticoid receptor antagonist and SGLT2 inhibitor.

#240 · Cardiac Diseases

Which beta blockers have mortality benefit in HFrEF?

Answer

Carvedilol, metoprolol succinate and bisoprolol.

#241 · Cardiac Diseases

When is ivabradine used?

Answer

Sinus rhythm at 70 beats per minute or above on maximally tolerated beta blocker, with HFrEF.

#242 · Cardiac Diseases

How long before elective surgery should SGLT2 inhibitors be stopped?

Answer

3 to 4 days, to reduce the risk of euglycemic ketoacidosis.

#243 · Cardiac Diseases

Why hold ACE inhibitors or ARBs before cardiac surgery?

Answer

They are associated with vasoplegia after bypass, so many centers hold them 24 hours beforehand.

#244 · Cardiac Diseases

Indications for cardiac resynchronization therapy?

Answer

Ejection fraction 35 percent or less, sinus rhythm, LBBB with QRS 150 ms or more, and NYHA II to IV symptoms on medical therapy.

#245 · Cardiac Diseases

Primary-prevention ICD criteria?

Answer

Ejection fraction 35 percent or less, at least 40 days after MI or 90 days after revascularization, on 3 months of therapy, with survival over 1 year expected.

#246 · Cardiac Diseases

What perioperative step is needed for an ICD during electrocautery?

Answer

Suspend tachytherapies by magnet or programming, and apply external defibrillator pads.

#247 · Cardiac Diseases

UNOS status 1 heart transplant criteria include which supports?

Answer

VA ECMO, non-dischargeable surgically implanted biventricular support, and a durable device with life-threatening ventricular arrhythmia.

#248 · Cardiac Diseases

Which supports qualify for UNOS status 2?

Answer

Intra-aortic balloon pump, percutaneous endovascular device, or sustained ventricular tachycardia.

#249 · Cardiac Diseases

Resting heart rate of a denervated transplanted heart?

Answer

About 90 to 110 beats per minute, without vagal tone.

#250 · Cardiac Diseases

Which bradycardia treatments work in a transplant recipient?

Answer

Isoproterenol, epinephrine and pacing; atropine, glycopyrrolate and carotid massage do not.

#251 · Cardiac Diseases

Why is ephedrine less effective after heart transplant?

Answer

It acts partly indirectly through norepinephrine release from nerve terminals, which are denervated.

#252 · Cardiac Diseases

Triple immunosuppression after heart transplant?

Answer

A calcineurin inhibitor, an antimetabolite such as mycophenolate, and a corticosteroid.

#253 · Cardiac Diseases

Toxicities of calcineurin inhibitors?

Answer

Nephrotoxicity, hypertension, hyperkalemia, hypomagnesemia and neurotoxicity.

#254 · Cardiac Diseases

Define primary graft dysfunction.

Answer

Left, right or biventricular failure within 24 hours of transplant without another identifiable cause.

#255 · Cardiac Diseases

Late killer after heart transplant and why it is silent?

Answer

Cardiac allograft vasculopathy, a diffuse concentric coronary disease, silent because of denervation.

#256 · Cardiac Diseases

Which factor determines whether a pericardial effusion causes tamponade?

Answer

The speed of accumulation more than the volume.

#257 · Cardiac Diseases

What is Beck triad?

Answer

Hypotension, jugular venous distension and muffled heart sounds.

#258 · Cardiac Diseases

Define pulsus paradoxus.

Answer

Inspiratory fall in systolic arterial pressure greater than 10 mmHg.

#259 · Cardiac Diseases

CVP waveform change in tamponade?

Answer

Preserved x descent with loss of the y descent.

#260 · Cardiac Diseases

Hemodynamic finding in tamponade?

Answer

Equalization of diastolic pressures (CVP, PA diastolic and wedge) within about 5 mmHg.

#261 · Cardiac Diseases

Echo findings of tamponade?

Answer

Right atrial systolic collapse, right ventricular diastolic collapse, plethoric IVC and exaggerated respiratory variation of inflow velocities.

#262 · Cardiac Diseases

Why is TEE needed after cardiac surgery with suspected tamponade?

Answer

Loculated posterior clot can compress the atria without a large effusion visible on transthoracic views.

#263 · Cardiac Diseases

Anesthetic goals for tamponade before decompression?

Answer

Preserve preload, high heart rate, contractility and vascular resistance; avoid positive pressure ventilation until the pericardium is opened.

#264 · Cardiac Diseases

Preferred induction drug in tamponade?

Answer

Ketamine, with the surgeon scrubbed and ready.

#265 · Cardiac Diseases

Most common cause of constrictive pericarditis worldwide?

Answer

Tuberculosis.

#266 · Cardiac Diseases

Pressure tracing in constriction?

Answer

Dip-and-plateau (square root) ventricular pressure with a prominent y descent.

#267 · Cardiac Diseases

What is Kussmaul sign?

Answer

A rise in CVP on inspiration, seen in constriction.

#268 · Cardiac Diseases

How does tissue Doppler distinguish constriction from restriction?

Answer

Constriction has preserved or increased e' prime; restriction has low e' prime.

#269 · Cardiac Diseases

Which phenomenon marks constriction on echo?

Answer

Ventricular interdependence with septal bounce and mitral E velocity falling more than 25 percent with inspiration.

#270 · Cardiac Diseases

Determinants of myocardial oxygen demand?

Answer

Heart rate, wall tension (preload and afterload) and contractility.

#271 · Cardiac Diseases

What is left ventricular coronary perfusion pressure?

Answer

Aortic diastolic pressure minus left ventricular end-diastolic pressure.

#272 · Cardiac Diseases

Why is tachycardia harmful in ischemic heart disease?

Answer

It raises demand and shortens diastole, which cuts left coronary flow.

#273 · Cardiac Diseases

Which myocardial layer is most vulnerable to ischemia?

Answer

The subendocardium.

#274 · Cardiac Diseases

ECG criteria for STEMI?

Answer

ST elevation of 1 mm or more in two contiguous leads, 2 mm or more in V2 and V3 in men over 40.

#275 · Cardiac Diseases

Why is nitroglycerin cautioned in inferior MI?

Answer

Right ventricular involvement makes the patient preload dependent and prone to hypotension.

#276 · Cardiac Diseases

First-medical-contact-to-device goal for primary PCI?

Answer

Within 90 minutes of first medical contact; fibrinolysis if PCI cannot be done within 120 minutes.

#277 · Cardiac Diseases

Treatment of no-reflow after PCI?

Answer

Intracoronary adenosine, nitroprusside or verapamil.

#278 · Cardiac Diseases

Minimum delay for elective surgery after a drug-eluting stent?

Answer

6 months, and 3 months may be acceptable if the risk of delay is greater; bare-metal stents need 1 month.

#279 · Cardiac Diseases

Which CABG indications improve survival over medical therapy?

Answer

Left main stenosis of 50 percent or more, three-vessel disease especially with reduced ejection fraction, and proximal LAD with multivessel disease.

#280 · Cardiac Diseases

Which conduit has the best long-term patency in CABG?

Answer

The left internal thoracic artery to the LAD, patent above 90 percent at 10 years.

#281 · Cardiac Diseases

Why is a vasodilator used with radial artery grafts?

Answer

Radial grafts are prone to vasospasm, so calcium channel blocker or nitroglycerin is often given.

#282 · Cardiac Diseases

Hemodynamic problems of off-pump grafting to posterior vessels?

Answer

Heart displacement causes hypotension, mitral regurgitation and right ventricular compression; treat with Trendelenburg, volume and vasopressors.

#283 · Cardiac Diseases

Most common primary cardiac tumor in adults?

Answer

Myxoma, usually left atrial and attached to the fossa ovalis.

#284 · Cardiac Diseases

Valvular effects of carcinoid heart disease?

Answer

Right-sided fibrosis causing tricuspid regurgitation and pulmonary stenosis.

#285 · Cardiac Diseases

Anesthetic caution in cardiac amyloidosis?

Answer

Calcium channel blockers and digoxin bind amyloid fibrils and are poorly tolerated, as are vasodilators.

#286 · Cardiac Diseases

Key rhythm risks in cardiac sarcoidosis?

Answer

Heart block and ventricular tachycardia, so keep pacing and defibrillation ready.

#287 · Cardiac Diseases

Which storage disease has deficient alpha-galactosidase A?

Answer

Fabry disease, causing concentric hypertrophy and conduction disease.

#288 · Cardiac Diseases

Most common primary malignant cardiac tumor?

Answer

Angiosarcoma; metastases are more common than any primary cardiac tumor.

#289 · Cardiac Diseases

Most common pediatric cardiac tumor and its association?

Answer

Rhabdomyoma, associated with tuberous sclerosis.

#290 · Cardiac Diseases

When is a lateral or Trendelenburg position hazardous in left atrial myxoma?

Answer

When it lets the tumor obstruct mitral inflow, causing sudden hemodynamic collapse; use TEE and avoid tumor manipulation.

#291 · Valvular Heart Disease

Severe aortic stenosis by velocity, mean gradient and AVA?

Answer

Peak velocity 4 m/s or more, mean gradient 40 mmHg or more, AVA 1.0 cm² or less (indexed 0.6 cm²/m² or less).

#292 · Valvular Heart Disease

What is the simplified Bernoulli equation?

Answer

Peak gradient = $4 \times V^2$. A 4 m/s jet gives 64 mmHg.

#293 · Valvular Heart Disease

What is the dimensionless index and its severe AS cutoff?

Answer

LVOT VTI divided by AV VTI. A value of 0.25 or less supports severe AS and avoids the diameter error.

#294 · Valvular Heart Disease

Define stage D2 aortic stenosis.

Answer

Low-flow, low-gradient severe AS with reduced LVEF: AVA 1.0 cm² or less, velocity under 4 m/s, mean gradient under 40 mmHg.

#295 · Valvular Heart Disease

How does dobutamine stress echo separate true from pseudo-severe AS?

Answer

In true severe AS the gradient rises while the AVA stays small. In pseudo-severe AS the AVA rises as flow increases.

#296 · Valvular Heart Disease

Define paradoxical low-flow, low-gradient AS (stage D3).

Answer

Preserved LVEF with a stroke volume index under 35 mL/m², typically a small hypertrophied LV in an older woman.

#297 · Valvular Heart Disease

Why does AS patient tolerate tachycardia poorly?

Answer

Shortened diastole cuts coronary supply while wall tension and ejection time raise demand, causing ischemia even with clean coronaries.

#298 · Valvular Heart Disease

Why do patients with AS depend on atrial contraction?

Answer

The stiff hypertrophied LV gets up to 40 percent of filling from atrial kick, so losing sinus rhythm can cause collapse.

#299 · Valvular Heart Disease

What is Heyde syndrome?

Answer

Severe AS with gastrointestinal angiodysplasia bleeding due to acquired von Willebrand syndrome from loss of large multimers by high shear.

#300 · Valvular Heart Disease

Indications for AVR in aortic stenosis?

Answer

Symptomatic severe AS, asymptomatic severe AS with LVEF under 50 percent, or severe AS found during other cardiac surgery.

#301 · Valvular Heart Disease

Age guidance for SAVR versus TAVR?

Answer

SAVR favored under 65, TAVR favored over 80 or with limited life expectancy, shared heart team decision from 65 to 80.

#302 · Valvular Heart Disease

Hemodynamic goals for anesthesia in severe AS?

Answer

Sinus rhythm at 60 to 80 beats/min, adequate preload, preserved SVR and contractility. Treat hypotension with phenylephrine or norepinephrine.

#303 · Valvular Heart Disease

Why place external defibrillator pads before induction in severe AS?

Answer

CPR generates poor coronary flow across a stenotic valve, so early defibrillation is the main rescue.

#304 · Valvular Heart Disease

What is rapid ventricular pacing used for in TAVR?

Answer

Pacing at about 180 to 200 beats/min drops cardiac output to near zero for stable balloon valvuloplasty or valve deployment.

#305 · Valvular Heart Disease

Most common conduction complication after TAVR?

Answer

Complete heart block and new left bundle branch block, more frequent with self-expanding valves and a short membranous septum.

#306 · Valvular Heart Disease

List major TAVR complications.

Answer

Conduction block, paravalvular leak, annular rupture, coronary obstruction, valve embolization, dissection, stroke, vascular injury and tamponade.

#307 · Valvular Heart Disease

A patient becomes hypotensive after TAVR deployment. What is the first imaging step?

Answer

Examine echo at once for new pericardial effusion, new regurgitation or coronary obstruction.

#308 · Valvular Heart Disease

Common causes of chronic aortic regurgitation?

Answer

Aortic root dilation (Marfan, annuloaortic ectasia, hypertension, aortitis) and leaflet disease (bicuspid valve, rheumatic).

#309 · Valvular Heart Disease

Common causes of acute aortic regurgitation?

Answer

Infective endocarditis, type A aortic dissection and trauma.

#310 · Valvular Heart Disease

Echo criteria for severe AR?

Answer

Vena contracta over 0.6 cm, regurgitant volume 60 mL or more, regurgitant fraction 50 percent or more, EROA 0.30 cm² or more, PHT under 200 ms.

#311 · Valvular Heart Disease

Which Doppler flow finding supports severe AR?

Answer

Holodiastolic flow reversal in the descending thoracic aorta, more specific if seen in the abdominal aorta.

#312 · Valvular Heart Disease

Arterial waveform in chronic severe AR?

Answer

Wide pulse pressure, low diastolic pressure and rapid upstroke and collapse (water-hammer pulse).

#313 · Valvular Heart Disease

Why is acute AR worse tolerated than chronic AR?

Answer

The LV has not dilated, so LVEDP rises sharply. Premature mitral closure may appear on M-mode and the pulse pressure is not wide.

#314 · Valvular Heart Disease

Why avoid beta-blockers in acute severe AR?

Answer

They block the compensatory tachycardia that keeps diastole short and cardiac output up.

#315 · Valvular Heart Disease

Surgery triggers for asymptomatic severe AR?

Answer

LVEF 55 percent or less, or LV end-systolic dimension over 50 mm (indexed over 25 mm/m²).

#316 · Valvular Heart Disease

Anesthetic goals in severe AR?

Answer

Heart rate 80 to 100 beats/min, maintained preload and lowered afterload. Avoid bradycardia and increased SVR.

#317 · Valvular Heart Disease

Why is an intra-aortic balloon pump contraindicated in significant AR?

Answer

Diastolic augmentation pushes more blood back through the incompetent valve, increasing regurgitation.

#318 · Valvular Heart Disease

Why use an LV vent in AR on bypass?

Answer

Fibrillation or cooling before cross-clamping distends the LV, which is poorly tolerated. A vent decompresses it.

#319 · Valvular Heart Disease

How should cardioplegia be given in severe AR after aortic cross-clamp?

Answer

Directly into the coronary ostia after aortotomy or retrograde via the coronary sinus, since root delivery leaks into the LV.

#320 · Valvular Heart Disease

Main cause of mitral stenosis worldwide?

Answer

Rheumatic heart disease, with commissural fusion, leaflet thickening and a doming, hockey-stick anterior leaflet.

#321 · Valvular Heart Disease

Non-rheumatic causes of mitral stenosis?

Answer

Degenerative mitral annular calcification, radiation, congenital MS, carcinoid, lupus and obstructing masses.

#322 · Valvular Heart Disease

Define severe and very severe mitral stenosis by MVA.

Answer

Severe is 1.5 cm² or less. Very severe is 1.0 cm² or less. Normal is 4 to 6 cm².

#323 · Valvular Heart Disease

Pressure half-time formula for MVA?

Answer

MVA = 220 / PHT. A PHT of 220 ms gives 1.0 cm².

#324 · Valvular Heart Disease

Why must the heart rate be reported with a mitral gradient?

Answer

The mean gradient depends on flow and diastolic time, so tachycardia raises it without a change in valve area.

#325 · Valvular Heart Disease

Pressure tracing finding in MS with sinus rhythm versus atrial fibrillation?

Answer

The PCWP tracing shows a large a wave in sinus rhythm, which is lost in atrial fibrillation.

#326 · Valvular Heart Disease

Anticoagulant for AF with moderate to severe rheumatic MS?

Answer

Warfarin, a vitamin K antagonist. Direct oral anticoagulants are not used.

#327 · Valvular Heart Disease

Ideal candidate for percutaneous mitral balloon commissurotomy?

Answer

Symptomatic severe rheumatic MS with Wilkins score 8 or less, no left atrial thrombus and no more than mild MR.

#328 · Valvular Heart Disease

Heart rate goal and why in mitral stenosis?

Answer

Slow, about 60 to 80 beats/min, because longer diastole lowers the transmitral gradient and left atrial pressure.

#329 · Valvular Heart Disease

Which anesthetic factors raise PVR and worsen MS with pulmonary hypertension?

Answer

Hypoxia, hypercarbia, acidosis, nitrous oxide and light anesthesia.

#330 · Valvular Heart Disease

Main hazard of anesthesia in mitral stenosis?

Answer

Right ventricular failure from pulmonary hypertension, plus pulmonary edema with tachycardia or excess fluid.

#331 · Valvular Heart Disease

Describe Carpentier type I and II MR.

Answer

Type I is normal leaflet motion (annular dilation, perforation, cleft). Type II is excess motion (prolapse, flail, papillary rupture).

#332 · Valvular Heart Disease

Describe Carpentier type IIIa and IIIb MR.

Answer

IIIa is restricted motion in systole and diastole (rheumatic). IIIb is restricted in systole only (ischemic or functional tethering).

#333 · Valvular Heart Disease

Primary versus secondary mitral regurgitation?

Answer

Primary is a disease of the valve apparatus. Secondary (functional) is a disease of the ventricle with tethering or annular dilation.

#334 · Valvular Heart Disease

Why is LVEF overestimated in chronic severe MR?

Answer

Part of the stroke volume ejects into the low-pressure left atrium, so an LVEF of 60 percent or less is already abnormal.

#335 · Valvular Heart Disease

Why does acute MR cause pulmonary edema and shock?

Answer

A normal-sized, noncompliant atrium faces sudden volume, giving a large v wave and high left atrial pressure.

#336 · Valvular Heart Disease

Which papillary muscle ruptures more often with myocardial infarction and why?

Answer

The posteromedial muscle, because it has a single blood supply from the posterior descending artery.

#337 · Valvular Heart Disease

Echo criteria for severe primary MR?

Answer

Vena contracta 0.7 cm or more, EROA 0.40 cm² or more, regurgitant volume 60 mL or more, regurgitant fraction 50 percent or more.

#338 · Valvular Heart Disease

What does a large v wave on the PCWP trace suggest, and what limits it?

Answer

Possible severe MR, but it is also seen with a stiff atrium or VSD, so it is not specific alone.

#339 · Valvular Heart Disease

Indications for surgery in primary MR?

Answer

Symptomatic severe MR, or asymptomatic severe MR with LVEF 60 percent or less or LV end-systolic dimension 40 mm or more.

#340 · Valvular Heart Disease

Why is repair preferred over replacement for degenerative MR?

Answer

It preserves LV function and avoids lifelong anticoagulation. Replacement should spare the subvalvular apparatus.

#341 · Valvular Heart Disease

COAPT criteria for TEER in secondary MR?

Answer

Symptomatic on optimal therapy, LVEF 20 to 50 percent, LV end-systolic dimension 70 mm or less, PA systolic pressure 70 mmHg or less.

#342 · Valvular Heart Disease

What gradient after TEER suggests iatrogenic mitral stenosis?

Answer

A mean transmitral gradient above about 5 mmHg.

#343 · Valvular Heart Disease

Anesthetic goals in severe MR?

Answer

Heart rate 80 to 100 beats/min, maintained preload and reduced afterload. Avoid bradycardia and increased SVR.

#344 · Valvular Heart Disease

What is SAM after mitral repair?

Answer

Systolic anterior motion of the anterior leaflet obstructing the LVOT, causing hypotension and new MR.

#345 · Valvular Heart Disease

Risk factors for SAM?

Answer

Redundant posterior leaflet, small hyperdynamic LV, basal septal bulge and long anterior leaflet.

#346 · Valvular Heart Disease

Treatment of SAM after repair?

Answer

Slow the heart rate, give volume, raise afterload with phenylephrine, use a beta-blocker and stop inotropes. Return to bypass if unstable.

#347 · Valvular Heart Disease

Why can LV function drop after mitral repair?

Answer

Closing the low-pressure escape into the atrium raises afterload, unmasking a hidden LV dysfunction.

#348 · Valvular Heart Disease

Primary and secondary causes of tricuspid regurgitation?

Answer

Primary: Ebstein, rheumatic, carcinoid, endocarditis, lead damage. Secondary: annular dilation from RV dilation, pulmonary hypertension, left-sided disease or AF.

#349 · Valvular Heart Disease

Echo criteria for severe TR?

Answer

Vena contracta 0.7 cm or more, EROA 0.40 cm² or more, hepatic vein systolic flow reversal, dense triangular CW Doppler signal.

#350 · Valvular Heart Disease

CVP waveform changes in severe TR?

Answer

Loss of the x descent and a prominent v wave merging with the c wave (ventricularization).

#351 · Valvular Heart Disease

Why is thermodilution cardiac output unreliable in severe TR?

Answer

Indicator regurgitates back into the right atrium, distorting the curve.

#352 · Valvular Heart Disease

When is tricuspid annuloplasty added to left-sided valve surgery?

Answer

For severe TR, and reasonable for mild to moderate TR with annular dilation above 40 mm or 21 mm/m².

#353 · Valvular Heart Disease

Anesthetic goals in severe TR?

Answer

Maintain preload, support the RV, avoid increased PVR and keep a normal heart rate.

#354 · Valvular Heart Disease

Hemodynamic findings in tricuspid stenosis?

Answer

Large a wave in the CVP, slow y descent and a mean gradient of 5 mmHg or more. It is usually rheumatic with mitral disease.

#355 · Valvular Heart Disease

INR goal for a mechanical bileaflet aortic valve without risk factors?

Answer

2.5 (range 2.0 to 3.0).

#356 · Valvular Heart Disease

INR goal for a mechanical mitral valve?

Answer

3.0 (range 2.5 to 3.5).

#357 · Valvular Heart Disease

Which anticoagulants are contraindicated with a mechanical valve?

Answer

Direct oral anticoagulants. Warfarin is required.

#358 · Valvular Heart Disease

How are patient-prosthesis mismatch grades defined?

Answer

Indexed EOA $0.85 \text{ cm}^2/\text{m}^2$ or less is moderate and under $0.65 \text{ cm}^2/\text{m}^2$ is severe.

#359 · Valvular Heart Disease

Echo clue to prosthetic aortic valve obstruction?

Answer

A high gradient, a dimensionless index below 0.25 and a prolonged acceleration time.

#360 · Valvular Heart Disease

What causes hemolytic anemia after valve replacement?

Answer

Usually a paravalvular leak, with high LDH, low haptoglobin and schistocytes.

#361 · Valvular Heart Disease

Duke criteria major findings in endocarditis?

Answer

Positive blood cultures with a typical organism, and evidence of endocardial involvement (vegetation, abscess, new dehiscence or regurgitation).

#362 · Valvular Heart Disease

Which organism most often causes endocarditis in injection drug users?

Answer

Staphylococcus aureus, most often on the tricuspid valve.

#363 · Valvular Heart Disease

When is surgery indicated for endocarditis?

Answer

Heart failure from valve dysfunction, uncontrolled infection (abscess, fistula, persistent bacteremia), fungal organisms, or vegetation over 10 mm with embolism.

#364 · Valvular Heart Disease

What does new PR prolongation or heart block signal in endocarditis?

Answer

Perivalvular, usually aortic root, abscess extending into the conduction system.

#365 · Valvular Heart Disease

Timing of cardiac surgery after intracranial hemorrhage in endocarditis?

Answer

Delay about 4 weeks when possible, unless refractory heart failure or uncontrolled sepsis forces earlier surgery.

#366 · Adult Congenital Heart Disease

What defines Eisenmenger syndrome?

Answer

A large unrepaired left-to-right shunt causes obstructive pulmonary vascular disease. PVR exceeds SVR, the shunt reverses to right-to-left, and the patient becomes cyanotic.

#367 · Adult Congenital Heart Disease

What is the central hemodynamic goal in a patient with Eisenmenger syndrome under anesthesia?

Answer

Keep SVR from falling and PVR from rising. Use phenylephrine or vasopressin and avoid hypoxia, hypercarbia, acidosis and vasodilators.

#368 · Adult Congenital Heart Disease

Why is closing a defect contraindicated in fixed Eisenmenger physiology?

Answer

The right ventricle cannot handle full afterload from fixed high PVR, and the shunt acts as a pop-off for the right ventricle.

#369 · Adult Congenital Heart Disease

What precaution applies to every IV line in a patient with a possible right-to-left shunt?

Answer

Use air filters and remove all bubbles, because paradoxical air or particulate embolism can reach the brain.

#370 · Adult Congenital Heart Disease

How does Fontan circulation produce pulmonary blood flow?

Answer

Systemic venous return flows passively to the pulmonary arteries with no subpulmonary pump. Flow depends on CVP, low PVR and low intrathoracic pressure.

#371 · Adult Congenital Heart Disease

What ventilation strategy suits a Fontan patient?

Answer

Use the lowest mean airway pressure with short inspiratory times and low PEEP, and return to spontaneous breathing early. Positive pressure reduces venous return and raises PVR.

#372 · Adult Congenital Heart Disease

What is the first response to low output in a Fontan patient?

Answer

Give volume and lower PVR with oxygen, correction of acidosis and inhaled nitric oxide, then add inotropes if ventricular function is poor.

#373 · Adult Congenital Heart Disease

Why does a fenestrated Fontan need air filters?

Answer

The fenestration is a right-to-left communication between the baffle and the atrium, which allows paradoxical embolism.

#374 · Adult Congenital Heart Disease

What preparation reduces risk before a redo sternotomy?

Answer

Review CT for structures behind the sternum, secure large-bore access, have blood ready, place defibrillator pads, and consider peripheral cannulation and bypass before re-entry.

#375 · Adult Congenital Heart Disease

What peak Doppler gradient defines severe pulmonary stenosis?

Answer

A peak gradient above 64 mmHg, which is a velocity above 4 m/s. Moderate is 36 to 64 mmHg and mild is below 36 mmHg.

#376 · Adult Congenital Heart Disease

Which first-line intervention treats valvar pulmonary stenosis in most patients?

Answer

Balloon valvuloplasty for a mobile, domed valve.

#377 · Adult Congenital Heart Disease

What serious risk is tested during transcatheter pulmonary valve placement in a conduit?

Answer

Coronary compression, tested by balloon inflation with simultaneous coronary angiography or aortography.

#378 · Adult Congenital Heart Disease

Which indications support pulmonary valve replacement for asymptomatic severe PR?

Answer

RV end-diastolic volume index above about 160 mL/m², end-systolic volume index above about 80 mL/m², falling RV function, sustained arrhythmia, or QRS above 180 ms.

#379 · Adult Congenital Heart Disease

Where should the arterial line be placed in an adult with coarctation repair planned?

Answer

In the right radial artery, since the left subclavian may be included in the repair or arise below the obstruction. A lower-extremity line shows the true gradient.

#380 · Adult Congenital Heart Disease

What gradient generally supports intervention for coarctation?

Answer

An upper-to-lower extremity gradient above 20 mmHg, or lower if imaging shows significant narrowing with collateral flow.

#381 · Adult Congenital Heart Disease

What are the key features of Ebstein anomaly?

Answer

Apical displacement of the septal and posterior tricuspid leaflets, atrialized right ventricle, tricuspid regurgitation, a frequent interatrial shunt and accessory pathways such as WPW.

#382 · Adult Congenital Heart Disease

How should atrial fibrillation with pre-excitation in Ebstein anomaly be treated?

Answer

Use procainamide or cardioversion. AV nodal blockers can accelerate conduction over the accessory pathway.

#383 · Adult Congenital Heart Disease

What are the four components of tetralogy of Fallot?

Answer

A large malaligned VSD, right ventricular outflow tract obstruction, an overriding aorta and right ventricular hypertrophy.

#384 · Adult Congenital Heart Disease

How is a hypercyanotic tet spell treated?

Answer

Give 100 percent oxygen and volume, reduce agitation, raise SVR with phenylephrine and use a beta-blocker such as esmolol. Avoid inotropes and vasodilators.

#385 · Adult Congenital Heart Disease

Which ventricle supports the systemic circulation after a Mustard or Senning atrial switch?

Answer

The morphologic right ventricle, which risks late failure, tricuspid regurgitation, atrial arrhythmias and baffle obstruction or leak.

#386 · Thoracic Aortic Disease

Which aortic dissection types are Stanford type A and type B?

Answer

Type A involves the ascending aorta regardless of tear site. Type B does not involve the ascending aorta and arises distal to the left subclavian artery.

#387 · Thoracic Aortic Disease

How do DeBakey types I, II, and III differ?

Answer

Type I starts in the ascending aorta and extends to at least the arch. Type II is confined to the ascending aorta. Type III begins distal to the left subclavian; IIIa stays above the diaphragm, IIIb extends below it.

#388 · Thoracic Aortic Disease

What is an intramural hematoma?

Answer

Bleeding into the aortic media from the vasa vasorum with no intimal flap and no flow in the false channel. Ascending IMH is generally managed like type A dissection.

#389 · Thoracic Aortic Disease

What is a penetrating atherosclerotic ulcer?

Answer

An atherosclerotic plaque that ulcerates through the internal elastic lamina into the media, most often in the descending thoracic aorta of an older patient.

#390 · Thoracic Aortic Disease

What is the leading cause of death in acute type A dissection?

Answer

Rupture into the pericardium causing cardiac tamponade.

#391 · Thoracic Aortic Disease

Which coronary artery is most often involved in type A dissection?

Answer

The right coronary artery, producing inferior ST elevation from flap extension into its ostium.

#392 · Thoracic Aortic Disease

What is the first-line imaging test in a stable patient with suspected dissection?

Answer

Computed tomography angiography, which defines the tear, extent, branch involvement, and false lumen thrombus.

#393 · Thoracic Aortic Disease

Which test is preferred for a hemodynamically unstable patient with suspected dissection?

Answer

Transesophageal echocardiography, done in the operating room, with sensitivity near 98 percent.

#394 · Thoracic Aortic Disease

What is the TEE blind spot in aortic imaging?

Answer

The distal ascending aorta and proximal arch, because the air-filled trachea and left main bronchus lie between the probe and the aorta.

#395 • Thoracic Aortic Disease

How do the true and false lumens look on TEE?

Answer

The true lumen is usually smaller and expands in systole with systolic color flow. The false lumen is larger, with spontaneous echo contrast or thrombus.

#396 • Thoracic Aortic Disease

What are the 2022 ACC/AHA hemodynamic targets in acute aortic dissection?

Answer

Heart rate 60 to 80 beats per minute and systolic pressure below 120 mmHg, while maintaining end-organ perfusion.

#397 · Thoracic Aortic Disease

Why must a beta blocker precede a vasodilator in dissection?

Answer

A vasodilator alone causes reflex tachycardia and raises dP/dt , which can propagate the dissection. Beta blockade lowers heart rate and dP/dt first.

#398 · Thoracic Aortic Disease

How is uncomplicated type B dissection managed?

Answer

Medically, with anti-impulse therapy. Complicated type B (rupture, malperfusion, refractory pain, rapid expansion) is treated with endovascular or open repair.

#399 · Thoracic Aortic Disease

Why is right axillary cannulation favored for arch surgery?

Answer

It gives antegrade flow, allows selective antegrade cerebral perfusion by clamping the innominate artery, and avoids retrograde atheroma embolization.

#400 · Thoracic Aortic Disease

Where should the arterial line go with right axillary cannulation?

Answer

Use the left radial or a femoral artery for systemic pressure. A right radial line reads falsely high and shows cerebral perfusion pressure during selective antegrade perfusion.

#401 · Thoracic Aortic Disease

What signs suggest the arterial cannula is in the false lumen?

Answer

High line pressure, rapid loss of reservoir volume, falling radial pressure, or false lumen filling on TEE. Stop and reposition at once.

#402 · Thoracic Aortic Disease

Why should the pericardium not be drained before bypass in dissection with tamponade?

Answer

Decompression can raise aortic wall pressure and cause rupture. Establish bypass first.

#403 · Thoracic Aortic Disease

What activated clotting time is targeted for cardiopulmonary bypass in aortic surgery?

Answer

Above 480 seconds after full heparinization.

#404 · Thoracic Aortic Disease

How much does cerebral metabolic rate fall per degree Celsius of cooling?

Answer

About 6 to 7 percent per degree.

#405 · Thoracic Aortic Disease

What nasopharyngeal temperature range defines moderate hypothermia for circulatory arrest?

Answer

20.1 to 28 degrees Celsius, commonly used with selective antegrade cerebral perfusion.

#406 · Thoracic Aortic Disease

What temperature gradients are limited during cooling and rewarming?

Answer

Keep the water to blood gradient at no more than 10 degrees, and keep venous outflow temperature below 37 degrees during rewarming.

#407 · Thoracic Aortic Disease

What is the typical flow and pressure goal for selective antegrade cerebral perfusion?

Answer

About 10 mL/kg/min, titrated to a right radial pressure of 50 to 70 mmHg.

#408 · Thoracic Aortic Disease

What central venous pressure limit applies to retrograde cerebral perfusion?

Answer

Keep superior vena cava pressure below 25 mmHg.

#409 · Thoracic Aortic Disease

Alpha-stat versus pH-stat: how do they differ?

Answer

Alpha-stat leaves the gas uncorrected for temperature and preserves autoregulation. pH-stat adds carbon dioxide to hold corrected pH at 7.40, raising cerebral flow but abolishing autoregulation.

#410 · Thoracic Aortic Disease

What is a pseudoaneurysm of the aorta?

Answer

A contained rupture whose wall is only adventitia or surrounding tissue and clot, not all three layers. It needs urgent repair.

#411 · Thoracic Aortic Disease

Which genetic syndrome dissects at a smaller diameter than Marfan?

Answer

Loeys-Dietz syndrome (TGFB1 or TGFB2), with repair at 4.0 to 4.5 cm depending on genotype.

#412 · Thoracic Aortic Disease

What is the surgical size threshold for a sporadic ascending aortic aneurysm?

Answer

5.5 cm, or 5.0 cm at an experienced center in a low-risk patient with risk factors.

#413 · Thoracic Aortic Disease

At what size is the ascending aorta replaced during other cardiac surgery?

Answer

4.5 cm or larger when the patient is already undergoing aortic valve or other cardiac surgery.

#414 · Thoracic Aortic Disease

What growth rate prompts earlier ascending aneurysm repair?

Answer

0.3 cm per year over two consecutive years, or 0.5 cm in a single year.

#415 · Thoracic Aortic Disease

Which Crawford extent carries the highest risk of paraplegia?

Answer

Extent II, from the left subclavian to below the renal arteries.

#416 · Thoracic Aortic Disease

What aneurysm sizes trigger descending thoracic repair?

Answer

5.5 cm or larger when TEVAR is feasible, and 6.0 cm or larger for open repair.

#417 · Thoracic Aortic Disease

What is the formula for spinal cord perfusion pressure?

Answer

Mean arterial pressure minus cerebrospinal fluid pressure, or minus central venous pressure if that is higher.

#418 · Thoracic Aortic Disease

What CSF drainage practice protects the spinal cord?

Answer

Keep CSF pressure about 10 mmHg or lower, limit drainage to roughly 10 to 15 mL per hour, and watch for subdural hematoma from rapid drainage.

#419 · Thoracic Aortic Disease

Which anesthetic conditions preserve motor evoked potentials?

Answer

Total intravenous technique, volatile agent below about 0.5 MAC, and no neuromuscular blockade after induction.

#420 · Thoracic Aortic Disease

Where does blunt traumatic aortic injury most often occur?

Answer

At the aortic isthmus just distal to the left subclavian artery, near the ligamentum arteriosum.

#421 · Electrophysiology Procedures

Initial treatment of an unstable tachyarrhythmia?

Answer

Synchronized cardioversion, not a drug trial.
Pulseless VF or VT gets unsynchronized defibrillation and CPR.

#422 · Electrophysiology Procedures

Typical biphasic energy for AF, flutter, and monomorphic VT with a pulse?

Answer

AF 120 to 200 J, flutter 50 to 100 J, monomorphic VT with a pulse 100 J, all synchronized.

#423 · Electrophysiology Procedures

Adenosine dosing for regular narrow complex tachycardia?

Answer

6 mg rapid IV push, then 12 mg if needed, through a proximal vein with a flush, after a vagal maneuver.

#424 · Electrophysiology Procedures

What drugs change the effect of adenosine?

Answer

Theophylline and caffeine blunt it; dipyridamole and carbamazepine exaggerate it. In a transplanted heart it causes prolonged asystole, so reduce the dose.

#425 · Electrophysiology Procedures

Atrial fibrillation with WPW and a wide irregular fast rate: what to avoid and what to use?

Answer

Avoid AV nodal blockers (adenosine, beta blockers, calcium channel blockers, digoxin). Use synchronized cardioversion if unstable, or procainamide or ibutilide if stable.

#426 · Electrophysiology Procedures

Treatment of torsades de pointes?

Answer

Magnesium sulfate 2 g IV, potassium above 4.5 mEq/L, stop QT-prolonging drugs, and overdrive pacing or isoproterenol for pause-dependent torsades. Defibrillate if pulseless.

#427 · Electrophysiology Procedures

Atropine dosing and its limits in bradycardia?

Answer

1 mg IV, repeat to a maximum of 3 mg. It is unreliable in Mobitz II, infranodal third degree block, and the denervated transplanted heart.

#428 · Electrophysiology Procedures

Why is complete heart block common after aortic valve surgery?

Answer

The AV conduction tissue lies next to the aortic annulus. Epicardial pacing wires are first line.

#429 · Electrophysiology Procedures

Primary prevention ICD criteria?

Answer

LVEF 35 percent or less, NYHA II or III, at least 40 days after MI and 90 days after revascularization, on guideline-directed therapy, with survival expected beyond one year.

#430 · Electrophysiology Procedures

Secondary prevention ICD indication?

Answer

Survivors of VF or hemodynamically unstable VT that is not caused by a reversible cause.

#431 · Electrophysiology Procedures

Strongest CRT indication?

Answer

LVEF 35 percent or less, sinus rhythm, LBBB with QRS 150 ms or longer, and NYHA II to ambulatory IV symptoms on guideline-directed therapy.

#432 · Electrophysiology Procedures

Where is the CRT left ventricular lead placed and what is its main implant risk?

Answer

Through the coronary sinus into a posterolateral vein. Coronary sinus dissection or perforation is the recognized risk.

#433 · Electrophysiology Procedures

What is conduction system pacing and why use it?

Answer

His bundle or left bundle branch area pacing captures the native conduction system for narrow activation and helps avoid pacing-induced cardiomyopathy.

#434 · Electrophysiology Procedures

What does a subcutaneous ICD lack?

Answer

Chronic bradycardia pacing, antitachycardia pacing, and CRT. It gives only brief backup pacing after a shock.

#435 · Electrophysiology Procedures

Who is a good S-ICD candidate and who is not?

Answer

Good: young patients, poor venous access, prior device infection. Not: patients needing pacing or with monomorphic VT that responds to ATP.

#436 · Electrophysiology Procedures

Main implant risk of a leadless pacemaker?

Answer

Cardiac perforation with tamponade, because it is placed in the right ventricle through a large femoral venous sheath.

#437 · Electrophysiology Procedures

Decode the NBG pacing code positions.

Answer

I chamber paced, II chamber sensed, III response to sensing (I, T, D, O), IV rate modulation (R).

#438 · Electrophysiology Procedures

What is pacemaker syndrome?

Answer

Loss of AV synchrony, usually from VVI pacing with intact sinus function, causing cannon A waves, hypotension, and fatigue.

#439 · Electrophysiology Procedures

List causes of failure to capture.

Answer

High threshold from hyperkalemia, acidosis, ischemia, or drugs such as flecainide, and lead dislodgement.

#440 · Electrophysiology Procedures

Define oversensing and its danger.

Answer

The device senses noise such as cautery or myopotentials as native beats and inhibits pacing, which is dangerous in a pacemaker-dependent patient.

#441 · Electrophysiology Procedures

**How does a magnet break
pacemaker-mediated tachycardia?**

Answer

It converts the device to asynchronous pacing, which interrupts the retrograde sensing loop.

#442 · Electrophysiology Procedures

Why turn off rate response for surgery?

Answer

Minute ventilation sensors react to ventilator or impedance monitoring and accelerometers to vibration, causing inappropriate fast pacing.

#443 · Electrophysiology Procedures

What does a magnet do to a pacemaker?

Answer

Usually asynchronous pacing at a fixed rate that depends on the manufacturer and battery state, only while the magnet is in place.

#444 · Electrophysiology Procedures

What does a magnet do to an ICD?

Answer

Suspends tachycardia detection and therapy in most devices. Pacing continues in its programmed mode, and it never causes asynchronous pacing.

#445 · Electrophysiology Procedures

Perioperative plan for a pacemaker-dependent patient having monopolar cautery above the umbilicus?

Answer

Program asynchronous mode or use a magnet, use bipolar cautery or short bursts, and monitor perfusion by plethysmography or an arterial line.

#446 · Electrophysiology Procedures

Pad placement near a CIED generator?

Answer

Keep pad edges at least 8 cm from the generator, and place ICD defibrillation pads before draping.

#447 · Electrophysiology Procedures

When is a CIED interrogation required after surgery?

Answer

Before leaving a monitored setting if the device was reprogrammed, major EMI occurred, or cardioversion was delivered.

#448 · Electrophysiology Procedures

Catastrophic complication of transvenous lead extraction and preparation for it?

Answer

SVC or right atrial tear with tamponade. Have a surgeon, perfusion, large bore IV, arterial line, pads, blood in the room, and echocardiography.

#449 · Electrophysiology Procedures

Key anesthetic needs for AF ablation?

Answer

General anesthesia, heparin keeping ACT above 300 seconds, and vigilance for tamponade, volume overload from irrigated catheters, and phrenic or esophageal injury.

#450 · Electrophysiology Procedures

What is the anesthetic signature of pulsed field ablation?

Answer

Strong skeletal muscle contraction with each pulse train, needing deep neuromuscular blockade, plus risk of coronary spasm and hemolysis.

#451 · Electrophysiology Procedures

Presentation of atrioesophageal fistula?

Answer

Fever, chest pain, and embolic neurologic events 1 to 4 weeks after ablation. Avoid endoscopy with air insufflation and diagnose with chest CT.

#452 · Electrophysiology Procedures

What are the anesthetic concerns for VT ablation?

Answer

Poor reserve, unstable VT during mapping, sedation that may suppress inducibility, possible Impella or ECMO support, and tamponade with epicardial access.

#453 · Electrophysiology Procedures

PASS criteria after LAA occluder deployment?

Answer

Position, Anchor (tug test), Size (compression about 8 to 20 percent), Seal (residual leak under 5 mm).

#454 · Electrophysiology Procedures

Anticoagulation rule before elective cardioversion of AF lasting 48 hours or more?

Answer

At least 3 weeks of therapeutic anticoagulation or TEE to exclude LAA thrombus, then at least 4 weeks afterward.

#455 · Electrophysiology Procedures

ALARA and the effect of distance on radiation dose?

Answer

As low as reasonably achievable using time, distance, and shielding. Doubling distance cuts dose to one quarter.

#456 · Postoperative Considerations

What tidal volume and plateau pressure define lung-protective ventilation in ARDS?

Answer

6 mL/kg of predicted body weight with plateau pressure at or below 30 cm H₂O. Aim for a driving pressure (plateau minus PEEP) under 15 cm H₂O.

#457 · Postoperative Considerations

How is predicted body weight calculated for a man?

Answer

$50 + 2.3 \times (\text{height in inches minus } 60) \text{ kg}$. For a woman it is $45.5 + 2.3 \times (\text{height in inches minus } 60) \text{ kg}$.

#458 · Postoperative Considerations

What are the Berlin ARDS severity bands by PaO₂/FiO₂ on at least 5 cm H₂O PEEP?

Answer

Mild 200 to 300, moderate 100 to 200, severe 100 or less.

#459 · Postoperative Considerations

Which ARDS interventions have mortality evidence in moderate to severe disease?

Answer

Lung-protective ventilation and prone positioning for at least 12 to 16 hours daily started early. Inhaled pulmonary vasodilators improve oxygenation but not survival.

#460 · Postoperative Considerations

How do TRALI and TACO differ?

Answer

TRALI: hypotension or normal pressure, normal or low filling pressures, no diuretic response. TACO: hypertension, high BNP and filling pressures, diuretic response.

#461 · Postoperative Considerations

What is the time window and treatment for TRALI?

Answer

Noncardiogenic pulmonary edema within 6 hours of a plasma-containing product. Treatment is supportive and lung-protective; stop the transfusion, notify the blood bank, avoid diuretics if hypotensive.

#462 · Postoperative Considerations

What rapid shallow breathing index predicts weaning success?

Answer

Respiratory rate divided by tidal volume in liters below 105.

#463 · Postoperative Considerations

Why can a spontaneous breathing trial fail in a cardiac surgery patient?

Answer

Removing positive pressure raises LV preload and afterload and can cause pulmonary edema in a marginal LV. Rising wedge pressure or falling SvO₂ during the trial suggests a cardiac cause.

#464 · Postoperative Considerations

What is the usual timing and incidence of postoperative atrial fibrillation after cardiac surgery?

Answer

20 to 40 percent after isolated CABG and up to 50 percent after valve surgery, peaking on postoperative days 2 to 3.

#465 · Postoperative Considerations

What is the treatment of unstable postoperative atrial fibrillation?

Answer

Synchronized cardioversion. If stable, control rate with a beta blocker or diltiazem when LV function is preserved, or use amiodarone when LV function is depressed.

#466 · Postoperative Considerations

How does the cardiac surgery arrest algorithm treat VF or pulseless VT?

Answer

Up to three stacked shocks first, no routine 1 mg epinephrine boluses, emergency resternotomy within 5 minutes, and amiodarone 300 mg after the third failed shock.

#467 · Postoperative Considerations

Why is AV sequential pacing preferred over ventricular pacing after cardiac surgery?

Answer

It preserves atrial contribution to filling, raising cardiac output roughly 15 to 25 percent in a stiff, hypertrophied ventricle.

#468 · Postoperative Considerations

What findings suggest early graft dysfunction after CABG?

Answer

New ST elevation in a graft territory, new regional wall motion abnormality, ventricular arrhythmia, or unexplained hypotension with rising filling pressures. Troponin alone is not specific.

#469 · Postoperative Considerations

What hemodynamic pattern suggests postoperative tamponade?

Answer

Abrupt fall in chest tube drainage, rising and equalized CVP, PA diastolic and wedge pressures within about 5 mm Hg, narrow pulse pressure, low output and oliguria.

#470 · Postoperative Considerations

Why does transthoracic echo often miss postoperative tamponade?

Answer

Clots are frequently localized and posterior, compressing the left atrium or one chamber. TEE is far more sensitive.

#471 · Postoperative Considerations

What is the definitive treatment of unstable postoperative tamponade?

Answer

Surgical re-exploration, even at the bedside. Needle pericardiocentesis fails for clotted blood.

#472 · Postoperative Considerations

How is low cardiac output syndrome defined?

Answer

Cardiac index below 2.0 L/min/m² with hypoperfusion signs such as SvO₂ below 60 percent, rising lactate and oliguria.

#473 · Postoperative Considerations

What hemodynamic pattern defines postoperative RV failure?

Answer

High CVP with low or normal wedge pressure, CVP exceeding wedge pressure, a distended RV with septal flattening, and low output.

#474 · Postoperative Considerations

How is postoperative RV failure managed?

Answer

Lower RV afterload (avoid hypoxia, hypercapnia, acidosis, excess PEEP; inhaled nitric oxide or epoprostenol), support with milrinone or epinephrine, use vasopressin for systemic pressure, keep AV synchrony and avoid volume overload.

#475 · Postoperative Considerations

What defines vasoplegia after cardiopulmonary bypass?

Answer

Low SVR (below about 800 dyne-s/cm⁵), normal or high cardiac index, hypotension and normal filling pressures, poorly responsive to catecholamines.

#476 · Postoperative Considerations

What is the stepwise treatment of vasoplegia?

Answer

Norepinephrine, then vasopressin 0.03 to 0.04 units/min, then angiotensin II or methylene blue 1 to 2 mg/kg over 20 to 60 minutes. Hydroxocobalamin 5 g is a rescue.

#477 · Postoperative Considerations

When should methylene blue be avoided?

Answer

With serotonergic drugs (serotonin syndrome risk), G6PD deficiency, and pulmonary hypertension.

#478 · Postoperative Considerations

What viscoelastic findings point to residual heparin?

Answer

Prolonged R time or clotting time in the native cup that corrects in the heparinase cup. Treat with small protamine doses of 25 to 50 mg.

#479 · Postoperative Considerations

What are common chest tube criteria for re-exploration?

Answer

Above 400 mL in the first hour, above 300 mL/h for 2 hours, above 200 mL/h for 4 hours, or a sudden large output of more than 500 mL.

#480 · Postoperative Considerations

Why are NSAIDs avoided after CABG?

Answer

They raise the risk of graft failure, myocardial infarction and renal injury, and carry an FDA boxed warning in the CABG setting.

#481 · Other Topics

What does a p value of 0.03 mean?

Answer

If the null hypothesis were true, data at least this extreme would occur about 3 percent of the time. It is not the probability that the null hypothesis is true.

#482 · Other Topics

Event rate is 10 percent with control and 6 percent with treatment. What is the number needed to treat?

Answer

Absolute risk reduction is 4 percent, so NNT is 1 divided by 0.04, which is 25. Relative risk reduction is 40 percent.

#483 · Other Topics

Which test compares time to an event between two groups?

Answer

Kaplan-Meier curves compared with the log-rank test. Cox proportional hazards regression gives the hazard ratio.

#484 · Other Topics

How do prevalence changes affect predictive values?

Answer

Sensitivity and specificity do not change. When prevalence falls, positive predictive value falls and negative predictive value rises.

#485 · Other Topics

What is intention-to-treat analysis?

Answer

Patients are analyzed in the group they were randomized to, whatever treatment they received. It preserves the benefit of randomization.

#486 · Other Topics

Which test is used for two independent groups with skewed data?

Answer

The Mann-Whitney U test. The paired nonparametric equivalent is the Wilcoxon signed rank test.

#487 · Other Topics

What does the perioperative surgical home aim to do?

Answer

It is a patient-centered, team-led model that coordinates care from the decision for surgery through recovery and discharge.

#488 · Other Topics

What is alert fatigue in an electronic medical record?

Answer

Clinicians override warnings because too many are low value. Selective, well-designed decision support reduces it.

#489 · Other Topics

What is a sentinel event?

Answer

An unexpected occurrence involving death or serious physical or psychological injury, or the risk of it, that triggers an urgent review.

#490 · Other Topics

What does the Swiss cheese model say about harm?

Answer

Harm occurs when holes in several layers of defense line up. Most errors are system failures, not individual carelessness.

#491 · Other Topics

Which safety intervention is strongest: education, a policy, or a forcing function?

Answer

A forcing function, such as a connector that cannot be mismatched, because it removes the possibility of the error. Education and policy are weaker.

#492 · Other Topics

What is closed-loop communication?

Answer

The sender gives an order, the receiver repeats it back, and the sender confirms it was heard correctly.

#493 · Other Topics

What are the three stages of the WHO surgical safety checklist?

Answer

Sign in, time out and sign out. Any team member may stop the line during the time out.

#494 · Other Topics

What does PDSA stand for and what is its purpose?

Answer

Plan, Do, Study, Act. It is an iterative cycle for testing a small change and then adopting, adapting or abandoning it.

#495 · Other Topics

What are the three types of QI measures?

Answer

Outcome, process and balancing measures. A balancing measure detects unintended harm elsewhere in the system.

#496 · Other Topics

How do FMEA and root cause analysis differ?

Answer

FMEA is prospective and scores severity, occurrence and detectability into a risk priority number. Root cause analysis is retrospective, done after an event.

#497 · Other Topics

What are the 5S steps?

Answer

Sort, set in order, shine, standardize, sustain.

#498 · Other Topics

Why is risk adjustment needed in public reporting?

Answer

So that a center treating sicker patients is not unfairly penalized when outcomes are compared with other centers.

#499 · Other Topics

What is stereotype threat?

Answer

The drop in performance when a person fears confirming a negative stereotype about their own group.

#500 · Other Topics

Who should interpret for a patient with limited English during consent?

Answer

A qualified medical interpreter, not a family member or ad hoc staff member.